

Bansilal Ramnath Agarwal Charitable Trust's
VISHWAKARMA INSTITUTE OF TECHNOLOGY
PUNE-411037

Department of Engineering, Sciences and Humanities

Calculus and Statistics HS1076

UNIT V – DESCRIPTIVE STATISTICS

Home Assignment

For this assignment you will solve 3 questions.

Solve questions mentioned against your roll number from sheet given at the end

Date of submission: 31/01/2025

Q.1 The students in a math class took the Scholastic Aptitude Test. Their scores are shown below. 356, 640, 345, 349, 574, 348, 618, 581, 470, 482
Find the range of the data. Represent the data as a Histogram and box plot. State outliers, if any. Find mean and mode, Karl Pearson's coefficient of Skewness, Bowley's coefficient of Skewness, Coefficient of Kurtosis. Hence State the distribution is positively/ negatively skewed or is symmetric. Also state it is platykurtic/mesokurtic/leptokurtic.

Q.2 Exam scores for 100 randomly selected college students has the frequency distribution was shown in the table below.

Class	90 – 98	99 – 107	108 – 116	117 – 125	126 - 134
Frequency	5	20	40	30	5

Find the range of the data. Represent the data as a Histogram. Find mean and mode, Karl Pearson's coefficient of Skewness, Coefficient of Kurtosis. Hence State the distribution is positively/ negatively skewed or is symmetric. Also state it is platykurtic/mesokurtic/leptokurtic.

Q.3 The following is prices, rounded to the nearest cent, charged per gallon of standard unleaded gasoline in the San Francisco Bay area in June 1997.
3.88, 3.90, 3.93, 3.90, 3.93, 3.96, 3.88, 3.94, 3.96, 3.88, 3.94, 3.99, 3.98
Find the range of the data. Represent the data as a frequency table. Represent the data as a Histogram and box plot. State outliers, if any. Find mean and mode, Karl Pearson's coefficient of Skewness, Bowley's coefficient of Skewness, Coefficient of Kurtosis. Hence State the distribution is positively/ negatively skewed or is symmetric. Also state it is platykurtic/meso/kurtic/leptokurtic.

Q.4	The following data give noise levels measured at 36 different times directly outside of Grand Central Station in Manhattan. 82,89,94,110,74,122,112,95,100,78,65,60,90,83,87,75,114,85, 69,94,124, 115,107,88, 97,74, 72,68,83,91,90,102,77,125,108,65 Find the range of the data. Represent the data as a frequency table. Represent the data as a Histogram and box plot. State outliers, if any. Find mean and mode, Karl Pearson's coefficient of Skewness, Bowley's coefficient of Skewness, Coefficient of Kurtosis. Hence State the distribution is positively/ negatively skewed or is symmetric. Also state it is platykurtic/mesokurtic/leptokurtic.																				
Q.5	The following data is the marks of students in a certain examination. <table><tr><td>Marks</td><td>40 – 50</td><td>50 – 60</td><td>60 – 70</td><td>70 – 80</td><td>80 – 90</td><td>90 – 100</td></tr><tr><td>No. of Students</td><td>20</td><td>5</td><td>10</td><td>15</td><td>9</td><td>5</td></tr></table> Represent the data as a Histogram. Find mean and mode, Karl Pearson's coefficient of Skewness, Coefficient of Kurtosis. Hence State the distribution is positively/ negatively skewed or is symmetric. Also state it is platykurtic/mesokurtic/leptokurtic.							Marks	40 – 50	50 – 60	60 – 70	70 – 80	80 – 90	90 – 100	No. of Students	20	5	10	15	9	5
Marks	40 – 50	50 – 60	60 – 70	70 – 80	80 – 90	90 – 100															
No. of Students	20	5	10	15	9	5															
Q.6	The following data represent the number of vehicles passing per hour at certain junction point. 20, 30, 15, 20, 25, 10 Find the range of the data. Represent the data as a Histogram and box plot. State outliers, if any. Find mean and mode, Karl Pearson's coefficient of Skewness, Bowley's coefficient of Skewness, Coefficient of Kurtosis. Hence State the distribution is positively/ negatively skewed or is symmetric. Also state it is platykurtic/mesokurtic/leptokurtic.																				
Q.7	Consider the following data. <table><tr><td>Class</td><td>1 - 5</td><td>6 - 10</td><td>11 - 15</td><td>16 - 20</td><td>21 - 25</td><td>26 - 30</td></tr><tr><td>Frquency</td><td>5</td><td>13</td><td>10</td><td>8</td><td>8</td><td>6</td></tr></table> Represent the data as a Histogram. Find mean and mode, Karl Pearson's coefficient of Skewness, Coefficient of Kurtosis. Hence State the distribution is positively/ negatively skewed or is symmetric. Also state it is platykurtic/mesokurtic/leptokurtic.							Class	1 - 5	6 - 10	11 - 15	16 - 20	21 - 25	26 - 30	Frquency	5	13	10	8	8	6
Class	1 - 5	6 - 10	11 - 15	16 - 20	21 - 25	26 - 30															
Frquency	5	13	10	8	8	6															
Q.8	The following data give the numbers of hours studying by 10 randomly selected college students during the past week: 5, 7, 14, 0, 7, 9, 4, 10, 0, 8 Find the range of the data. Represent the data as a Histogram and box plot. State outliers, if any. Find mean and mode, Karl Pearson's coefficient of Skewness, Bowley's coefficient of Skewness, Coefficient of Kurtosis. Hence State the distribution is positively/ negatively skewed or is symmetric. Also state it is platykurtic/mesokurtic/leptokurtic.																				
Q.9	Consider the following data. <table><tr><td>Class</td><td>4 - 5</td><td>7 - 8</td><td>10 – 11</td><td>13 - 14</td><td>16 - 17</td></tr><tr><td>Frquency</td><td>5</td><td>9</td><td>14</td><td>7</td><td>5</td></tr></table> Represent the data as a Histogram. Find mean and mode, Karl Pearson's coefficient of Skewness, Coefficient of Kurtosis. Hence State the distribution is positively/ negatively skewed or is symmetric. Also state it is platykurtic/mesokurtic/leptokurtic.							Class	4 - 5	7 - 8	10 – 11	13 - 14	16 - 17	Frquency	5	9	14	7	5		
Class	4 - 5	7 - 8	10 – 11	13 - 14	16 - 17																
Frquency	5	9	14	7	5																
Q.10	A class of sixth grade students kept accurate records on the amount of time they spent playing video games during a one-week period. The times (in hours) are listed below. 26.7 14.7 8.3 12.9 15.1, 28.7 23.0 23.6 14.3,11.0 Find the range of the data. Represent the data as a Histogram and box plot. State outliers, if any. Find mean and mode, Karl Pearson's coefficient of Skewness, Bowley's coefficient of Skewness, Coefficient of Kurtosis. Hence State the distribution is positively/ negatively skewed or is symmetric. Also state it is																				

	platykurtic/mesokurtic/leptokurtic.
Q.11	The test scores of 19 students are listed below. 91,99,86,54,72,85,97,91,90,66,82,83,78,88,77,80,92,94,98 Represent the data as a Histogram and box plot. State outliers, if any. Find mean and mode, Karl Pearson's coefficient of Skewness, Bowley's coefficient of Skewness, Coefficient of Kurtosis. Hence State the distribution is positively/ negatively skewed or is symmetric. Also state it is platykurtic/mesokurtic/leptokurtic.
Q.12	The following data give the number of new cars sold at a dealership during a 20-day period. 8,5,12,3,9,10,6,12,8,8,4,16,10,11,7,7,3,5,9,11 Represent the data as a frequency table. Represent the data as a Histogram and box plot. State outliers, if any. Find mean and mode, Karl Pearson's coefficient of Skewness, Bowley's coefficient of Skewness, Coefficient of Kurtosis. Hence State the distribution is positively/ negatively skewed or is symmetric. Also state it is platykurtic/mesokurtic/leptokurtic.
Q.13	The nine measurements that follow are furnace temperatures recorded on successive batches in a semiconductor the manufacturing process (units are): 953, 950, 948, 955, 951, 949, 957, 954, 955. a) Calculate the sample mean, sample variance, and standard deviation. b) Find the median. c) Construct a box plot of the data. State outliers, if any.

Roll No	Q1	Q2	Q3
1	1	8	10
2	1	7	13
3	4	9	11
4	2	9	12
5	4	8	12
6	5	7	10
7	1	8	11
8	3	8	12
9	4	6	13
10	5	7	13
11	3	9	13
12	2	8	11
13	4	7	12
14	5	8	13
15	5	6	10
16	3	8	10
17	3	6	10
18	3	6	10
19	3	8	11
20	5	7	12
21	5	9	10
22	3	8	13

Roll No	Q1	Q2	Q3
23	1	8	12
24	3	6	13
25	1	8	13
26	5	7	12
27	4	6	10
28	5	8	13
29	4	8	13
30	4	8	13
31	3	8	12
32	5	7	11
33	1	8	12
34	2	8	11
35	3	7	13
36	1	7	11
37	1	7	12
38	1	6	12
39	4	8	11
40	5	6	10
41	1	7	10
42	1	7	11
43	4	9	13
44	5	6	10
45	3	6	10
46	4	8	13
47	4	8	12
48	5	7	11
49	3	8	12
50	3	6	11
51	5	9	13
52	4	9	13
53	4	6	11
54	1	8	11
55	4	9	12
56	4	8	13
57	4	6	11
58	4	7	10
59	5	8	13
60	3	9	13
61	2	9	12
62	4	6	10
63	3	9	11
64	5	6	10
65	3	9	12
66	5	7	12
67	1	7	11

Roll No	Q1	Q2	Q3
68	2	9	10
69	3	9	11
70	4	8	10
71	3	6	11
72	3	9	13
73	5	8	12
74	5	6	13
75	2	7	12
76	2	7	12
77	3	9	10
78	4	7	12
79	3	6	11
80	3	6	10
81	5	6	12